

RoomZen OS

The Smart Operating System for Student Housing

1. Executive Summary & Vision

RoomZen is an AI-powered student housing and hostel management SaaS platform designed to eliminate paper registers, unorganized WhatsApp complaint groups, spreadsheets, and manual room allocation. By providing digitized workflows for Students, Hostel Admins, Wardens, Repair Staff, and Gate Security Guards, RoomZen brings operating efficiency, safety, and financial transparency to campus housing.

2. Technology Stack & Architecture

- Framework: Next.js 16 (App Router with Turbopack & Server Components)
- Frontend Language & Styling: TypeScript, TailwindCSS v4, Framer Motion
- UI Components & Icons: Custom Primitives, Glassmorphism Cards, Lucide React
- Database & Backend: Supabase PostgreSQL, Row Level Security (RLS), Auth
- State Management: Zustand Unified Store with Hybrid Fallback State
- Analytics & Visuals: Recharts (Revenue, Occupancy, Complaint Triage)
- Form Validation & Utilities: React Hook Form, Zod, date-fns, qrcode.react, jsPDF

3. Multi-Role Dashboard Specifications

• Hostel Owner / Admin Dashboard

Real-time KPI metrics (Total Students, Bed Occupancy, Revenue, Pending Complaints), Recharts analytics area charts, quick room assignment, and notice creation.

• Student Portal & Mobile App

Room details, AI-assisted complaint submission, digital weekend gate pass requests with QR generation, mess menu schedule, and fee payment history.

• Chief Warden Control Desk

Gate pass approval queue with guardian contact phone call trigger, complaint triage & staff assignment, night attendance check sheet, and incident reports.

• Maintenance Staff Workspace

Work order job board (HVAC, Plumbing, Electrical, Wi-Fi), status transitions (Open -> In Progress -> Resolved), and repair completion photo uploads.

• Security Guard Gate Portal

Live QR Code pass scanner, student hash verifier, visitor check-in/check-out register, and gate exit timestamp logging.

4. Supabase Database Schema Architecture

- users (id, email, full_name, role, phone, avatar_url)
- students (id, user_id, student_id_number, room_id, emergency_contact, parent_name)
- rooms (id, room_number, floor, block, capacity, occupied, status, rent_amount)
- complaints (id, student_id, title, category, priority, description, status, assigned_staff_id)
- gate_passes (id, student_id, reason, destination, status, qr_code_hash, approved_by)
- payments (id, invoice_no, student_id, title, amount, status, paid_date, payment_method)
- visitors (id, student_id, visitor_name, visitor_phone, relation, time_in, time_out, status)
- mess_menu (id, day_of_week, meal_type, items, special_item, calories)
- announcements (id, title, content, priority, audience, created_by)
- chat_messages (id, sender_id, recipient_id, content, is_read, timestamp)

5. Git Team Branches & Repository Structure

- Repository: <https://github.com/adhithyapandiri-a11y/SyntaxSquad.git>
- Production Branch: main
- Teammate Branches: sravan, shivam, umair, aryan, nishpray